

Mini Cloud Deployment Platform - Project Brief

Project Name	Mini Cloud Deployment Platform
Date	2026
Project Client	Self-initiated portfolio project
Project Overview	This project demonstrates a reproducible cloud deployment platform using Terraform, Docker, GitHub Actions CI/CD, AWS ECS Fargate, an Application Load Balancer, and CloudWatch logging. The objective is to showcase infrastructure-as-code practices, automated deployments, container orchestration, and cost-aware cloud architecture.
Goals & Objectives	- Demonstrate Infrastructure as Code with Terraform - Deploy containerized workloads on AWS ECS Fargate - Implement CI/CD pipeline using GitHub Actions - Provide reproducible infrastructure using remote Terraform state in S3 - Maintain a cost-efficient architecture that can be destroyed and rebuilt on demand
Constraints & Assumptions	- Project remains beginner-friendly and reproducible - Expensive services such as RDS and NAT Gateway avoided - Infrastructure must be destroyable to keep costs near \$0 - Single ECS task required for MVP
Project Scope	- Python Flask demo application - Docker containerization - Terraform infrastructure provisioning - Amazon ECS Fargate service - Application Load Balancer with health checks - GitHub Actions CI/CD pipeline - CloudWatch logging and observability - Architecture diagram and documentation
Target Audience	- Recruiters evaluating cloud/platform engineering skills - Junior Cloud Engineer roles - DevOps and Platform Engineering teams
Success Criteria	- Infrastructure deploys successfully with terraform apply - Application reachable through ALB endpoint - CI/CD pipeline builds and deploys automatically - Infrastructure can be destroyed and rebuilt reliably - Documentation clearly explains architecture and deployment workflow
Cost Strategy	Infrastructure runs only when needed. After testing or demonstrations the environment is removed using terraform destroy so default AWS cost stays approximately \$0.
Remote Terraform State	Terraform state stored in Amazon S3 so infrastructure can be rebuilt from any machine and managed across multiple devices.

Architecture Diagram Requirement

Repository includes architecture diagram:

Developer -> GitHub -> GitHub Actions -> Docker -> ECR -> ECS Fargate
-> ALB -> Browser

Stored in docs/architecture-diagram.png and explained in docs/architecture.md.